Looking Away

Building 1: Moves

Arrows from a floor to itself, and what counting their "hows" gives.

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A *move* is an arrow from a floor to itself that is an inside and a picture (Chapter 0, W2). Forgetting *how* one thing moves to another gives a blur: Building 0. This book keeps the how. It reads Building 0 and is never read by it (the rule F7, between buildings).

Chapter numbers restart in this book; Building 0 is cited as "Building 0, Chapter n". The rules are Chapter 0 of Building 0.

Planned, not yet written (`plans/ALGEBRA-PATH.md`). Part II: Groups, the table; Cosets and Lagrange; Normal subgroups; Chapters 2, 7, 14 again over any ring; Modules and the Smith form; Actions; Counting and Sylow; Divisibility in any ring; Linear maps; Similarity; Forms; Field extensions; Finite fields; the Galois correspondence; Solvable ladders and radicals. Then the elliptic curves: the projective plane; plane curves; the group law as a blur; over $\mathbb{Z}/p$; the torus.

Part I. Moves on things

Chapter 1. Rearrangements

a bag of n things $\to$ its moves · from: Building 0, Chapters 0, 1, 3 · build, select, blur

Select the moves of n things. Glue all swaps together and exactly two bags are left: the sign appears. Glue a move to its relabellings and the bags are the shapes of its cycles.

The first chapter that keeps the *how*. A move is an arrow from a bag to itself that is an inside and a picture (Chapter 0, W2). Building 0 forgot how one thing was carried to another; here the carrying is the thing.

Given

G1 (Building 0, Chapter 0.) Build M1–M7; blur L1–L9; arrows W1–W4; W2: inside and picture.

G2 (Building 0, Chapter 3.) $\mathbb{Z}/2$ with $+$.

G3 (Building 0, Chapter 1.) Whole numbers; counting a finite bag.

Definitions

D1 (Build, M7; Select, M4.) Fix $n \geq 1$ and $[n] := \{1, \ldots, n\}$. S_n: the maps $\sigma : [n] \to [n]$ that are one-to-one (then also onto). Each is a *move* of $[n]$. $\sigma\tau := \sigma \circ \tau$ (do τ first). e: the move that does nothing.

D2 A *swap* $(i\,j)$ exchanges i and j and keeps the rest. A *cycle* $(i_1\, i_2 \ldots i_k)$ sends each i_r to i_{r+1} and i_k to i_1.

D3 (Inversions.) $\mathrm{inv}(\sigma)$: the number of pairs $i < j$ with $\sigma(i) > \sigma(j)$.

D4 (Relabelling.) For π in S_n: $\sigma^\pi := \pi\sigma\pi^{-1}$, the move σ with the things renamed by π.

Propositions: the floor

P1 (Laws, carried by W2.) $\circ$ is associative; e does nothing on either side; every move has an undo σ^{-1}. Not commutative for $n \geq 3$: $(1\,2)(2\,3) = (1\,2\,3)$ while $(2\,3)(1\,2) = (1\,3\,2)$.

P2 (Count.) S_n has $n!$ members.

P3 (Normal form: cycles.) Every move is a product of cycles on disjoint sets of things, in exactly one way up to the order of the cycles (follow each thing until it comes back).

P4 (Swaps are enough.) Every move is a product of swaps, by P3 and

$$(i_1 \ldots i_k) = (i_1\, i_2)(i_2\, i_3) \cdots (i_{k-1}\, i_k) \quad \text{(do the right-hand swap first).}$$

Blur 1: the sign

On S_n, $n \geq 2$. *O*: $\circ$.

B1 (From below.) Gluings: $s \sim t$ for any two swaps s, t.

B2 (From above.) Question: the parity of $\mathrm{inv}(\sigma)$, in $\mathbb{Z}/2$.

P5 (Survives: parity respects $\circ$.) $\mathrm{inv}(\sigma\tau) \equiv \mathrm{inv}(\sigma) + \mathrm{inv}(\tau) \pmod 2$. (For each pair $\{i, j\}$, $\sigma\tau$ turns it over exactly when one of τ on $\{i, j\}$ and σ on $\{\tau i, \tau j\}$ does; count the pairs mod 2.) A swap has an odd number of inversions: $(i\,j)$ with $i < j$ has $2(j - i) - 1$.

P6 (Appears.) The two blurs agree. There are exactly two bags, A_n (even) and its other bag (odd). (B1 puts every product of an even number of swaps with e and every odd product with $(1\,2)$, by P4; B2 tells them apart, by P5.) Name: $\mathrm{sgn}(\sigma) := (-1)^{\mathrm{inv}(\sigma)}$.

P7 (Dies: how many swaps.) $(1\,3) = (1\,2)(2\,3)(1\,2)$: one swap, or three. The number of swaps is not a question on moves; only its parity is.

P8 (Survives into Building 0.) $\mathrm{sgn} : S_n \to \{1, -1\}$ respects $\circ$ and $\times$: an arrow onto Building 0's $\mathbb{Z}/2$ (G2), written multiplicatively. By W4, $S_n \to \mathbb{Z}/2$ is the blur B1 followed by a matching.

Blur 2: relabelling

On S_n. *O*: nothing (L7: this blur respects neither $\circ$ nor its undo in general; say so).

B3 (From below.) Gluings: $\sigma \sim \sigma^\pi$ for every π.

B4 (From above.) Question: the *cycle type*, the list of cycle lengths of P3 in falling order (a partition of n).

P9 (Appears.) The two blurs agree. $(\pi(i_1 \dots i_k)\pi^{-1} = (\pi i_1 \dots \pi i_k)$: relabelling renames the cycles and keeps their lengths; two moves of the same type are relabellings of each other.) Normal form: one move per partition of n; the bags number $p(n)$: $1, 2, 3, 5, 7, 11$ for $n = 1, \dots, 6$.

P10 (Dies: $\circ$.) $(1\,2) \sim (2\,3)$, but $(1\,2)(1\,2) = e$ while $(2\,3)(1\,2) = (1\,3\,2)$.

P11 (Survives: sign and order.) $\mathrm{sgn}(\sigma) = (-1)^{n-c}$ with c the number of cycles (counting fixed things); the order of σ (least $m \geq 1$ with $\sigma^m = e$) is the lcm of the cycle lengths. Both are read off the type.

P12 (Bag sizes.) The bag of type $1^{m_1} 2^{m_2} \cdots$ has $n! \,/\, \prod_k k^{m_k}\, m_k!$ members. (Count the relabellings that give the same move.)

Theorems

T1 (Survey: every blur of S_n that respects $\circ$.) A blur of S_n respecting $\circ$ is fixed by its bag of e, which is a union of bags of Blur 2 closed under $\circ$. For $n \leq 6$ the survey is:
$n = 2$: two blurs. $n = 3$: three (e only; A_3; everything). $n = 4$: four (e; the three double swaps with e; A_4; everything). $n = 5$ and $n = 6$: three (e; A_n; everything).

Machine-checked (all unions of types closed under $\circ$). For $n \geq 5$ the classical theorem is that A_n glues to nothing smaller: A_n is *simple*. That is a count, proved in Part II.

T2 (The sign is the only arrow to $\mathbb{Z}/2$.) For $n \geq 2$, the arrows $S_n \to \{1, -1\}$ respecting $\circ$ are $\sigma \mapsto 1$ and sgn. (Relabelling glues all swaps (P9), so an arrow takes one value on them; P4.)

T3 (Every move system lives here. Cayley.) For any finite bag G of moves of something, closed under $\circ$ and undo, $g \mapsto (h \mapsto gh)$ is an inside of G into the moves of the bag G, respecting $\circ$. So every finite move system is found inside some S_n, $n =$ its size.

Remarks

R1 (What the blur did.) The sign is an Appears line: "glue all swaps" from below meets "parity of inversions" from above. That the bags are two and not one is the kept question P5, and it is the whole content of the classical theorem.

R2 (Building 0 seen from here.) The blur of Building 0, Chapter 3 ($n \sim 0$ on $\mathbb{Z}$) is "forget how far you moved" for the moves $x \mapsto x + k$. Every blur in Building 0 is such a forgetting; this book keeps the moves.

R3 (Old names.) S_n: the symmetric group. A_n: the alternating group. P3: disjoint cycle decomposition. P6: the sign is well defined. B3: conjugacy; its bags are conjugacy classes. P12: class sizes. T1: the normal subgroups of S_n. T3: Cayley's theorem.

R4 (Machine check, `checks/b1-ch1-check.py`.) P1, P2, P4 on all of S_n, $n \leq 6$; P5 for $n \leq 5$; P6 (closure of the swap gluing); P7; P9 and P12 (types and bag sizes); P11; T1 (survey); T2.

Exercises

E1 Write $\sigma = (1\,4\,2)(3\,5)$ in S_5 as swaps, and find $\operatorname{sgn}\sigma$ and the order of σ in two ways.

E2 Count the moves of type $2\,2\,1$ in S_5 by P12, and by listing.

E3 Show that in S_n the even moves are exactly the products of 3-cycles ($n \geq 3$).

E4 Which move of S_4 has the most inversions? How many does it have?

E5 Why is "σ and τ commute" not kept by Blur 2? Give a witness in S_3.

Chapter 2. Words

letters and their undos → the free move system · from: Building 0, Chapters 0, 1, 3; Chapter 1 · build, blur

Write words in letters and their undos, and glue a letter next to its undo to nothing. Each bag holds exactly one word with nothing left to cancel. Glue more, and every move system appears.

Chapter 1 started from moves already given. This chapter builds moves from nothing but names, then blurs. It is the build-then-blur shape of Building 0, Chapter 1, one level up.

Given

G1 (Building 0, Chapter 0, M3.) Words in a list of names; maps out of words by recursion.

G2 (Chapter 1.) Moves of a bag; $\circ$; P1.

G3 (Building 0, Chapters 1, 3.) $\mathbb{Z}$ and $\mathbb{Z}/n$ with $+$.

Definitions

D1 (Build, M3.) Fix a bag L of *letters*. For each letter s take a second name s^{-1}. $W(L)$: the finite strings in the names s, s^{-1}, with the empty string 1. Joining strings, $u \cdot v$, is the operation.

D2 A word is *reduced* if no s stands next to s^{-1} (in either order).

D3 (Each letter is a move of the reduced words.) For a reduced word r: $s * r := r$ with its first name removed if that name is s^{-1}, and sr otherwise; $s^{-1} * r$ in the same way. Each of these is a move (its undo is the other), so a word $u = x_1 \cdots x_m$ acts by $u * r := x_1 * (\cdots (x_m * r))$.

The blur

On $W(L)$. O: joining.

B1 (From below.) Gluings: $s\, s^{-1} \sim 1$ and $s^{-1} s \sim 1$, for each letter s.

B2 (From above.) Question: the move $r \mapsto u * r$ of the reduced words (D3).

Propositions

P1 (Appears.) The two blurs agree. (B1 keeps the question: $s * (s^{-1} * r) = r$. Every word is glued by B1 to a reduced one, by cancelling. Two reduced words $r \neq r'$ act differently: $r * 1 = r$.)

P2 (Normal form.) Each bag holds exactly one reduced word. (P1.) The bags, with joining, form a move system $F(L)$: the *free move system* on L. Undo: $(x_1 \cdots x_m)^{-1} = x_m^{-1} \cdots x_1^{-1}$.

P3 (Count.) With two letters, the reduced words of length $k \geq 1$ number $4 \cdot 3^{k-1}$: four choices for the first name, then three (anything but the undo of the one before).

P4 (Dies: length.) $s\,s^{-1}s \sim s$, of lengths 3 and 1.

P5 (Survives: length mod 2, and each letter's exponent sum.) Cancelling removes two names, one s and one s^{-1}. So $\ell(u)$ mod 2 and $\varepsilon_s(u) :=$ (number of s) − (number of s^{-1}) survive: arrows $F(L) \to \mathbb{Z}/2$ and $F(L) \to \mathbb{Z}$, respecting joining and $+$.

P6 (One letter.) $F(\{s\}) \to \mathbb{Z}$, $u \mapsto \varepsilon_s(u)$, is a matching. Building 0, Chapter 1 builds $\mathbb{Z}$ from pairs; here it is the free move system on one letter.

Theorems

T1 (Every move system is a picture of a free one.) Let G be a move system and choose g_s in G for each letter s. There is exactly one arrow $F(L) \to G$ respecting joining with $s \mapsto g_s$. (Recursion on words, G1; it respects B1 because $g_s g_s^{-1} = e$.) If the g_s generate G, the arrow is a picture, and by W4

$$G \text{ matches } F(L) \text{ blurred by the bag of words sent to } e.$$

T2 (Presentations: glue more.) For a list R of words, glue $r \sim 1$ for each r in R, respecting joining. Write the bags $\langle L \mid R \rangle$. Then the bag of 1 is the smallest bag that holds R, joins, undos, and relabellings uru^{-1} (as Chapter 1, T1: a zero bag must be kept by relabelling).

T3 (Old floors, as presentations.)
$\langle s \mid s^n \rangle$ matches $\mathbb{Z}/n$ (Building 0, Chapter 3).
$\langle s, t \mid sts^{-1}t^{-1} \rangle$ matches $\mathbb{Z}^2$: glue $st \sim ts$ and the normal form is $s^a t^b$ (the whole-number points of $\mathbb{R}^2$, Building 0, Chapter 6).
$\langle s, t \mid s^2, t^2, (st)^3 \rangle$ matches S_3 (Chapter 1), with $s = (1\,2)$, $t = (2\,3)$.
$\langle s, t \mid s^2, t^2, (st)^n \rangle$ has $2n$ members: the moves of a regular n-gon.
$\langle s, t \mid s^2, t^3, (st)^5 \rangle$ has 60 members.

T4 (Dies: deciding.) For some finite L and R, no procedure decides whether a word lies in the bag of 1 (classical, Novikov and Boone; reported). From below, a blur can be written down and still not be computable: Building 0 never met this, because its gluings always came with a normal form.

Remarks

R1 (Build, then blur, one level up.) Building 0, Chapter 1: pairs, glue $(a+1, b+1) \sim (a, b)$, normal form $a - b$. Here: words, glue $ss^{-1} \sim 1$, normal form the reduced word. The question that makes the blur from above (D3) is itself a move: in Building 1 even the questions are moves.

R2 (Where it goes.) T1 makes every move system a blur of a free one. Part II asks which blurs of a move system respect joining (cosets, normal bags), and counts bags.

R3 (Old names.) $F(L)$: the free group on L. D3 and P1: van der Waerden's proof of the normal form. P5: abelianisation to $\mathbb{Z}^L$ (exponent sums). T1: the universal property of free groups. T2: group presentations; the bag of 1 is the normal closure of R. T3: cyclic, free abelian, symmetric, dihedral groups; the last is A_5. T4: the word problem.

R4 (Machine check, `checks/b1-ch2-check.py`.) P1 on all words of length ≤ 8 in two letters; P3 to length 8; P5; T3 (orders by coset enumeration, and the arrow onto S_3).

Exercises

E1 Reduce $s\,t\,t^{-1}s^{-1}s\,t^{-1}\,t\,s$.

E2 Show that in $F(\{s,t\})$, $st \neq ts$, using the exponent sums or P2.

E3 In $\langle s,t \mid s^2, t^2, (st)^3\rangle$, show $sts = tst$.

E4 How many reduced words of length at most 3 are there in two letters?

E5 Why is "length is even" kept by B1 but "length is ≤ 5" not?